MyEcoPool

Driving Smart Urban Mobility

MGMT 59000

TEAM-9

Problem Statement

Urban commuters face trade-offs between travel time, cost, and environmental impact. Bus systems are affordable but often slow; private rides are fast but expensive and emit more CO_2.

Objective: Identify the optimal transportation strategy balancing efficiency, affordability, and sustainability to improve **MyEcoPool** operations and commuter experience.

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Dataset Overview

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Projected Impact (12 months)

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Strategic Alignment

MyEcoPool is not a standalone app. It's "Working Cities' Middle Inspector" and "Electrify the Fleet" objectives by providing:

1. A scalable KPI framework for accountability.
2. Centralized data driven decision **aligned coordination**.
3. Data driven insights allocation for micro-hubs.

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Key Insights

- Peak offer (Monday) 5:00 and 6:00 is 5.5 trips (Average 15.0).
- Peak demand (Friday) 17:00 is 19.0 trips (Average 15.0).
- Offer is consistently exceeds demand across all weekdays.
- Weekend offer underutilized. Revenue pooling can provide and offer 10% discount for weekend trips to increase ridership and utilization rate (Sat 12:00 is 10.0 vs 15.0 KPI).
- Average 1.4 Passengers per trip.

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THE KILLER FIGURE

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Dataset Overview

- Dataset: 60,000 trips representing Uber, Carpool, and Bus modes.
- Variables: Distance (miles), Time (minutes), Cost (USD), CO2 emissions (kg).
- Emission Factors: Bus (0.99), Carpool (2.68), Uber (32.82) kgs.
- Estimated by real-world benchmarks from:
  - Chicago Data Portal
  - Taxi and TNP datasets
  - Bureau of Transportation Statistics
  - EPA benchmarks data

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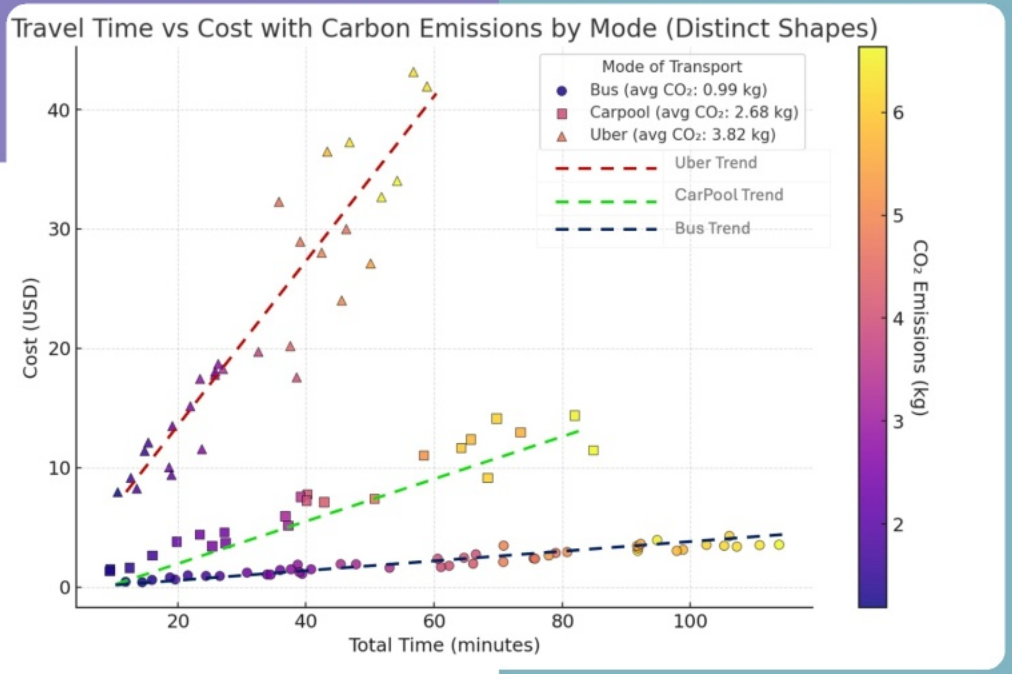
THE KILLER FIGURE

Travel Time vs Cost with Carbon Emissions



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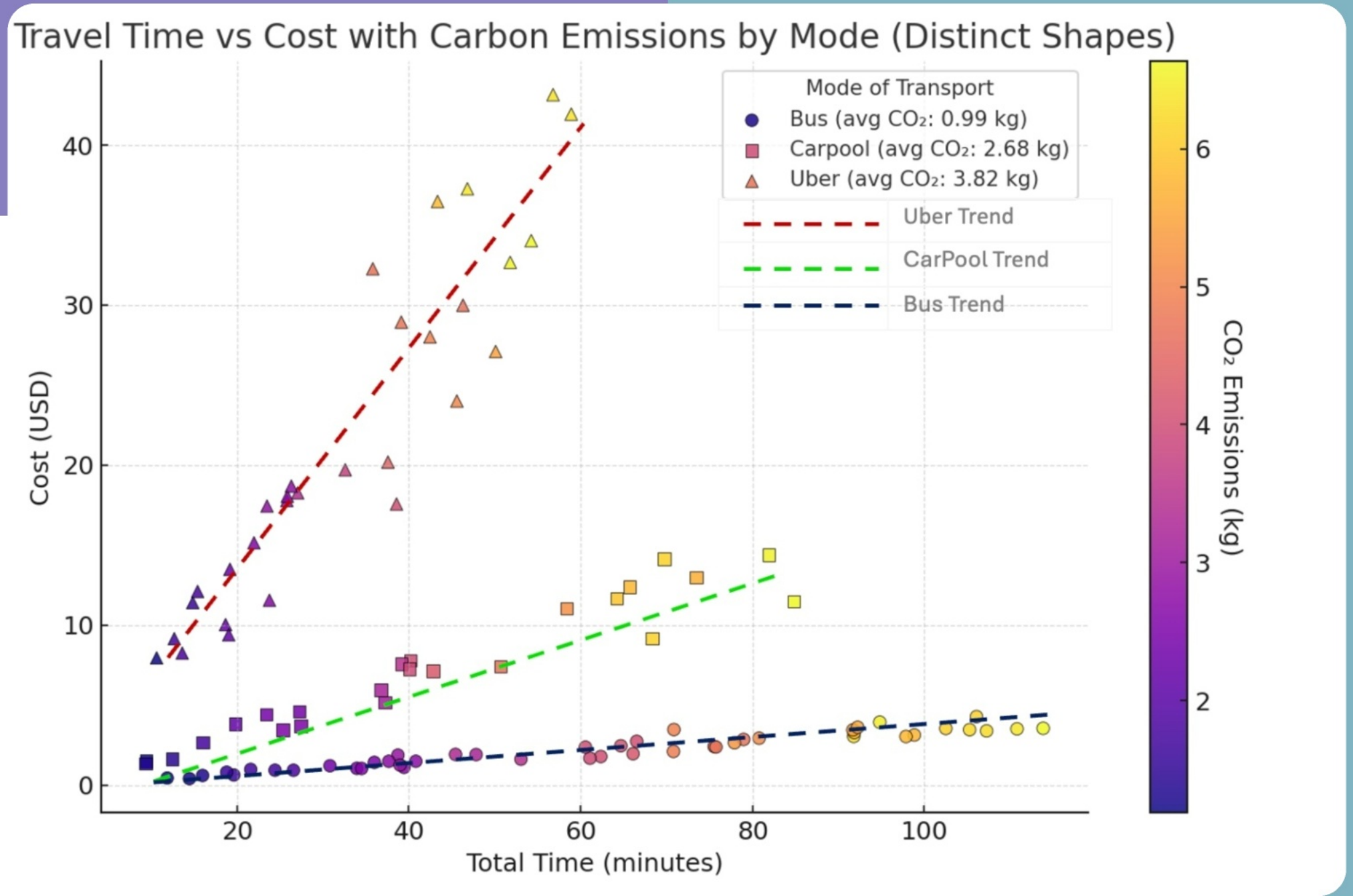
Travel Time vs Cost with Carbon Emissions



Recommendations & Business Case

1. Launch smart bus scheduling using real-time data analytics.

2. Introduce a **MyEcoPool** program integrated with bus routes.

3. Implement eco-driving training and route optimization.

Projected Impact (12 months)

- ~$22 million annual fuel + time savings.
- 100 000 metric tons CO2 avoided per year.
- 25% improvement in on-time bus performance (Smart Scheduling integration).

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Strategic Alignment

MyEcoPool directly advances CTA's "Mobility Data Modernization" and "Electrify the Fleet" objectives by providing:

1. A unified KPI framework for sustainability.
2. Scenario simulation for bus-carpool coordination.
3. Data-driven subsidy allocation for eco-rides.

Together, Let’s Make MyEcoPool the Heartbeat of a Greener City!

Empowering data-driven, sustainable, and efficient urban mobility.

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Q&A

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